2019 IEEE International Conference on Systems, Man and Cybernetics (SMC)
Bari, Italy. October 6-9, 2019

Target Classification in a Novel SSVEP-RSVP Based BCI Gaming System

Tapsya Nayak
Department of Electrical and Computer Engineer
The University of Texas at San Antonio
San Antonio, USA
ani254@my.utsa.edu

Li-Wei Ko
Department of Biological Science and Technology
National Chiao Tung University
Hsinchu, Taiwan
lwko@mail.nctu.edu.tw

Tzyy-Ping Jung
Swartz Center for Computational Neuroscience
University of California San Diego
San Diego, USA
jung@sccn.ucsd.edu

Yufei Huang
Department of Electrical and Computer Engineering
The University of Texas at San Antonio
San Antonio, TX
Yufei.Huang@utsa.edu

Abstract— **Recently game-based brain-computer interface (BCI) systems using electroencephalography (EEG) has been gaining popularity, providing a sophisticated experience to its users. Here we present such a novel hybrid system based on rapid serial visual presentation (RSVP) in conjunction with steady-state visual evoked potentials (SSVEP). Based on a matching computer game Jewel Quest a game is designed wherein a sequence of jewel images containing rare targets (< 3%) in an RSVP paradigm is presented on a display at four distinct locations each flickering at different rates (4, 5, 6 and 7 Hz). A score is awarded upon successful detection of target image from neural signals. During real-time implementation to achieve higher classification speeds, EEG signals were epoched at the onset of each image, creating a high degree of class overlap and imbalance. Given these challenges in our EEG datasets, we present classifiers that can classify single-trial EEG epochs at the onset of target image presentation accurately. Initial results from 14 subjects indicate Hidden Markov Model (HMM) with Dirichlet emission probabilities provide ~1% higher, on average, the area under the precision-recall curve (AUC-PR) compared to the ensemble technique Bagging, commonly used to handle class imbalance.**

Keywords—Brain-computer interface (BCI), steady-state visual evoked potential (SSVEP), rapid serial visual presentation (RSVP), Hidden Markov Model (HMM), Dirichlet distribution.

I. INTRODUCTION

Translation of human thought into real-world actions have been implemented successfully using brain-computer interface systems to achieve specific tasks by decoding stimuli based neural signals [1-5] . BCIs commonly rely on EEG to acquire these neural signals as they are inexpensive and non-invasive. Stimuli in BCI applications are generally designed to voluntarily generate a command to control external systems, for example, P300 based visual evoked potentials (VEPs) [6, 7], steady-state based VEPs [8-10] and sensorimotor (SMR) [11, 12] are used in BCI speller applications, control of wheelchairs and computer interfaces. These benefits have made them popular among individuals with disabilities and neurological conditions to interact independently in their surroundings. However, recent years has seen interest in BCI-based games (neurogaming) that attracts both individuals - disabled and healthy, as a new source of entertainment.

Past research groups have used well-known computer games in conjunction with BCI to challenge its users by

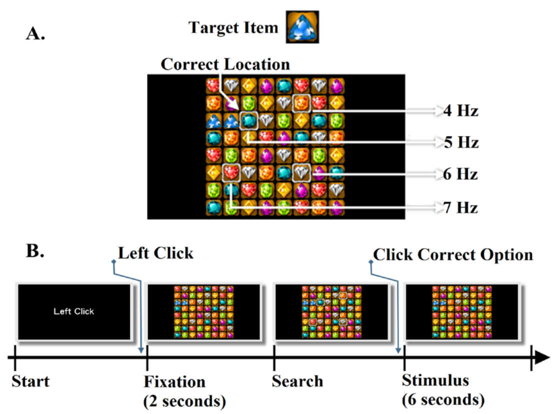

Figure 1. (A) BCI jewel game display. (B) Experiment timeline of each

providing a new control mechanism instead of joysticks, computer mouse or keyboard. For example, in [13] authors use the game Tetris in a hybrid BCI paradigm combining motor imagery (MI) and SSVEP to generate multiple commands; authors in [14] design a three-dimensional checker game board that employs P300 to move a character and in [15, 16] again use SSVEP signals to control the game to avoid obstacles. However, to achieve high online classification accuracies, especially to decode SSVEP frequencies, they either alter the game control paradigm or use longer EEG epochs ranging from one to three seconds with sliding window approach of 100-200 ms overlap. In addition to this, most hybrid BCI systems employ cascaded approach for classifications.

Here, we present a novel hybrid BCI game based on the popular computer game Jewel Quest that combines both VEPs i.e. based on P300s and SSVEP. In doing so, this provides an exciting experience to players in terms of visual perception. In this game setup, a player seated in front of a display unit is presented a matrix of jewel images with four white squares at different locations, each displaying a sequence of eight different jewels at four different flicker rates, see Fig. 1A. The player is instructed to identify and then focus on one of the white-bordered flickering locations where a jewel forms a matching triplet with its neighbors and is labeled as a 'Target' that occurs rarely. While the remaining non-matching jewels in the flickering sequence are labeled as 'Non-target'.

The goal of this system is to classify targets accurately at its onset in real-time without using a cascaded classification system for both VEPs. The classification here is challenging because single-trial EEG signals are epoched at the onset of each flickering image giving rise to a high degree of overlap among successive epochs, instead of a sliding window approach. An advantage of this approach is reduced time towards classifications, however achieving high classification accuracy is vital, especially since 'Target' class are sparse (<3%) to ensure the player is engaged. For the first time, to our knowledge, we attempt to address this issue in BCI systems using EEG data that focuses on classification accuracy with the class imbalance and overlap issues without comprising classification speed.

The remaining of the paper is organized as follows. Section II presents the game design, EEG data acquisition and preprocessing techniques and classification methods used. Lastly, section III presents the overall results obtained and concludes.

A. Game Design

Figure 1A, shows the monitor display of the jewel game designed using Unity. The game consists a 8x8 grid of jewel images with four white-bordered jewel images at different locations presenting a sequence of eight different jewel images in RSVP paradigm at flicker rates of 4, 5, 6 and 7 Hz, representing SSVEP frequencies, on a 100 Hz monitor refresh rate. Jewel images at each flicker location were presented at one half duty cycle and a black image on the other.

B. EEG Data Acquisition

EEG data were acquired using a Neuroscan system at a sampling rate of 500 Hz from a 32 channel system including two reference electrodes placed at left and right ear mastoids. Electrode impedance was maintained below 5 kΩ.

C. EEG Experiment

Fourteen healthy subjects (7 males) participated in the study with an average age of 24.8 (±3.31) years and all had a normal or corrected-to-normal vision. Each subject participated in seven sessions with 32 trials/session including 30 seconds of rest between each session. Fig. 1B illustrates the experimental timeline during a single trial. In each trial, the subject starts the game with a left mouse click, followed by fixation of 2 seconds, where the subject focuses his/her attention on the jewel matrix on the display. At the end of two seconds, 'search' phase begins where a white-border appears around four different locations with jewel images flickering in them. In this phase, the subject is instructed to identify the target location. Upon target location identification, the subject enters the next 'stimulus' phase with a left mouse click. In the stimulus phase, the subject focuses on pre-determined target location lasting for 6 seconds. In this phase, the target occurs only three times randomly. If the subject successfully identifies a target a score is awarded. Any time during 'search' or 'stimulus' phase if the subject believes the chosen target location is incorrect, he/she may cancel the trial by a right mouse click. In each trial, the target is associated with only one of the four flicker location (frequencies).

D. EEG Data Preprocessing and Epoching

EEG data acquired from stimulus phase from all subjects were used for classification purposes. EEG signals was first preprocessed by bandpass filtering between 0.5-50 Hz to remove any electrical noise and artefacts and re-referenced using averaged signals from mastoid electrodes. Following which, to mimic real-time classifications, one second EEG epochs were extracted at the onset of each image from the four white-bordered image locations. In the remaining of paper, this dataset will be referred to as sequential dataset. Beginning of each stimulus phase, images from all four locations are presented simultaneously, thus first EEG epoch extracted was removed. Also, for classification training purpose, if target image onset coincided with image onset from remaining three frequencies, the epoch is labeled as target. In this BCI game design, as SSVEP flicker frequencies are in close range (4-7 Hz) EEG epochs extracted suffer from the significant overlap between successive epochs with a minimum overlap of at least 2 ms. This may lead to misclassifications of epochs following target image as false-positives due to shared feature space among labels during classification.

The non-sequential dataset was also extracted wherein EEG epochs without any overlap for each class were selected. This dataset is balanced, i.e., the same number of epochs were extracted for each class.

E. Classification algorithm

The goal of the classifier is to detect the rare target epochs among non-targets from all four locations. Thus, classifiers were trained on five classes (four non-targets and one target) using supervised learning. During training, a classifier was first trained on the non-sequential dataset (balanced) using traditional classifiers (SVM and Bagging), wherein ~150 epochs from each label without any overlap were randomized and divided into five folds - 80% for training and rest for testing. Area under the receiving operating characteristic curve (AUC) metric is used to evaluate the performance of the classifier on the non-sequential dataset. The classifier model from each subject that provided the best AUC was chosen for sequential dataset classification.

For sequential dataset classification, since the target class is rare, the metric used to evaluate performance is the area under the precision-recall curve (AUC-PR). This dataset was evaluated on test data by dividing in two parts – first 80% of the sequential data for training and rest for testing.

Bagging using Decision Tree

Bagging (Bootstrap Aggregating) classification algorithm creates a classifier $\mathcal{H}$: D = [i], where i = [1, 2, 3, 4, 5] from the training set. The bagging method creates a sequence of classifiers H_m, m = 1,2… M representing the number of classifiers. These classifiers are combined into a compound classifier, whose prediction is given as weighted combinations of individual classifier predictions. This could be interpreted as a voting classifier as,

$$H(d_i) = \underset{m}{argmax}(\sum \alpha_m H_m(d_i)) \quad (1)$$

where α_m are weights such that the higher weights are given to classifiers that influence the final predictions greater than the other classifier models, which is determined during the training process.

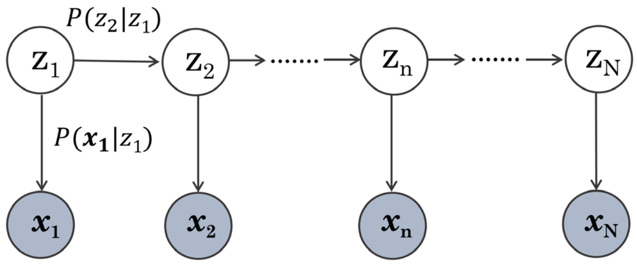

Figure 2. Graphical Model of first order HMM, white circles represent hidden states z_n and the shaded circles represent EEG observations $\boldsymbol{x}_n$.

Hidden Markov Model (HMM)

HMMs are probabilistic models commonly used in sequenced time series data. Fig. 2 shows the graphical model; to parameterize the model EEG observations are expressed as bagging scores computed for each EEG epoch are expressed as as $\boldsymbol{X}_{1:N} = \{\boldsymbol{x}_1 \ldots \boldsymbol{x}_N\}$, $\boldsymbol{x}_n \epsilon \mathbb{R}^{5x1}$ and N represents the number of EEG epochs extracted in a sequence. $\boldsymbol{Z}_{1:N}$ represents hidden states $z_n \epsilon \{1, 2, 3, 4, 5\}$, where state 5 represents target and states 1, 2, 3 and 4 represents nontargets from flicker frequencies 4, 5, 6 and 7 Hz. Given the HMM model parameters, i.e., initial probabilities (π_s), transition probability (A_{ks}) and emission probability (B_s), the goal of HMM is to compute the posterior probability of hidden state for a given time stamp given EEG observations,

$$p(z_k|x_{1:n}) = \underset{z}{\operatorname{argmax}}\, p(z_k|x_{1:n}) \tag{2}$$

Rewriting equation (2) at time k,

$$\mu_n(z_n) \,\alpha\, p(z_n|\, \boldsymbol{x}_{1:n})$$

The above equation can be written in a recursive form,

$$\mu_n(z_n) \,\alpha \sum_{z_{n-1}} p(z_n, z_{n-1}|\, x_{1:n})$$

$$\alpha \sum_{z_{n-1}} p(x_n|\, z_n, z_{n-1}, x_{1:n-1}) \,.\, p(z_n, z_{n-1}|\, x_{1:n-1})$$

$$\alpha \sum_{z_{n-1}} p(x_n|\, z_n) \,.\, p(z_n|\, z_{n-1}, x_{1:n-1})\, p(z_{n-1}|\, x_{1:n-1})$$

$$\alpha \sum_{z_{n-1}} p(x_n|\, z_n) \,.\, p(z_n|\, z_{n-1})\, p(z_{n-1}|\, x_{n-1})$$

$$\alpha\, p(x_n|\, z_n) \sum_{z_{n-1}} p(z_n|\, z_{n-1})\, p(z_{n-1}|\, x_{n-1})$$

where $n = 2, \ldots, N$ and $\mu_1(z_1) = p(z_1|\boldsymbol{x}_1) = p(z_1)p(\boldsymbol{x}_1|z_1)$.

The initial state probabilities (π_s), transition probabilities (A_{ks}), and emission probabilities (B_s) is determined during training from the sequenced dataset. As EEG observations $\boldsymbol{x}_n$ are Bagging scores representing softmax probabilities, the emission distribution is modeled using a Dirichlet distribution given by,

TABLE I. AVERAGE AUC OBTAINED FROM SVM AND BAGGING ON NON-SEQUENTIAL DATASET.

Classifier	***AUC***
SVM	0.7944
Bagging	0.8124

$$p(\boldsymbol{x}_n|z_n) = p(\boldsymbol{x}_n|\boldsymbol{\alpha})^{z_{ns}} = \left(\frac{\Gamma(\sum_k \alpha_k)}{\prod_k(\Gamma\alpha_k)} \prod_k x_k^{\alpha_k - 1}\right)^{z_{ns}} \tag{3}$$

where $k = \{1, 2, 3, 4, 5\}$, s represents hidden states $s \epsilon \{1, 2, 3, 4, 5\}$ and $\boldsymbol{\alpha}_s$ represents the parameters of Dirichlet distribution for the corresponding state s and were estimated using maximum likelihood estimates [17].

II. RESULTS AND DISCUSSION

In this section, we first present the performance of traditional classifiers SVM and Bagging on the non-sequential dataset. Next, we present the performance of the HMM filtering algorithm in comparison to Bagging results on sequential datasets.

A. Non-sequential EEG dataset

Classifier for the sequential dataset was chosen by first applying traditional classifiers on the non-sequential dataset. Two classifiers – SVM (linear kernel function) and Bagging were trained on EEG epochs using five-folds.

Table I shows the average AUC obtained from these two classifiers from all subjects. On comparing AUCs, Bagging outperforms SVM for all subjects by an average of 1.8%. Based on these results, Bagging classifier model from a non-sequential dataset from each subject is used to evaluate sequential dataset.

B. Sequential EEG dataset

Table II shows the performance of Bagging model obtained from the non-sequential dataset on the sequential dataset for all subject. The sequential dataset from each subject contains ~22,000 epochs with only ~650 target epochs. With-in subject sequential data analyses Bagging model the average AUC-PR from all subjects obtained is 0.4609. On further analyzing Bagging predictions, epochs successively following true-positives were misclassified as false-positives, thereby increasing the gaming scores.

To reduce the detection of false-positives, we implemented HMM that parametrizes EEG observations using Bagging scores with Dirichlet distribution as emission probabilities and furthermore HMMs take advantage of the information contained within transitional probabilities to compute posterior class probabilities. Specifically, while the

TABLE II. AUC-PR VALUES OBTAINED FROM BAGGING AND HMM ON SEQUENTIAL DATASETS.

AUC-PR	S01	S02	S03	S04	S05	S06	S07
Bagging	0.4072	0.408	0.6368	0.4449	0.5011	0.4074	0.4357
HMM	0.4211	0.432	0.6535	0.4825	0.5088	0.4277	0.4555
AUC-PR	**S08**	**S09**	**S10**	**S11**	**S12**	**S13**	**S14**
Bagging	0.4783	0.5214	0.4494	0.4315	0.4405	0.4174	0.4737
HMM	0.4893	0.5278	0.4643	0.4408	0.4619	0.4323	0.4903

overall goal of the classifier is to detect targets accurately, Bagging classifier is trained to detect true-targets accurately while HMM attempts to reduce the false-positives. The HMM model parameters were determined during training from the sequential dataset for each subject. From Table II, we observe that HMM consistently improves AUC-PR values for all subjects averaging at 0.4777, which is on average ~1% increase in AUC-PR compared to Bagging, observed due to the transitional probabilities that HMM employs. Specifically, we observe the highest increase in AUC-PR of 3.76% in subject 4 (see Fig 3, for precision-recall curves). For this BCI gaming application, reducing false-positives is vital as they incorrectly increase the score awarded to the player, which may be reduced by moving the threshold along the precision-recall curve. Additionally, on further assessing epoch predictions of both classifiers on the sequential dataset, we observe target detection missed by Bagging are mostly also missed by HMM. This may be attributed to either effect of attentional blinking arising from failure to observe the second of two targets presented relatively close together i.e. 250 ms apart [18] or to loss of attention/ fatigue experienced by the player.

In summary, for the first time, we present classification results of EEG epochs characterized with a high degree of class overlap and imbalance without using a cascaded approach and by implementing an EEG epoching method that utilizes less wait time. While further work is necessary, especially in BCI applications, to improve classification performance that ensures players' continuous engagement without frustration.

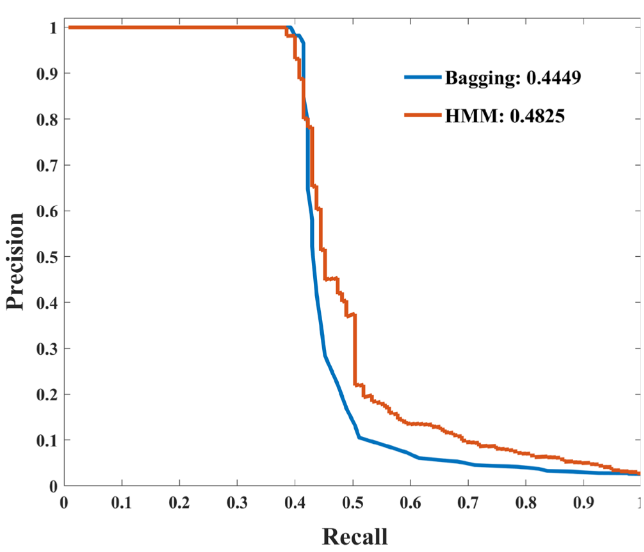

Figure 3. Subject 4 precision-recall curve using Bagging and HMM on sequential dataset, and its area under the curve (AUC-PR) values.

III. CONCLUSION

In conclusion, we present a novel hybrid BCI gaming application based on the computer game Jewel Quest where we attempt for the first time to improve rare target classifications characterized by class imbalance and overlap, that has not been addressed before in BCI applications to the best of our knowledge. Finally, further analysis needs to be conducted to improve performance using deep learning models for data with class imbalance and overlap that helps contribute to the BCI domain[19-21].

ACKNOWLEDGMENT

The research was sponsored by the Army Research Laboratory and was accomplished under Cooperative Agreement Number W911NF-10-2-0022. The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the official policies, either expressed or implies, of the Army Research Laboratory or the U.S. Government. The U.S. Government is authorized to reproduce and distribute reprints for the Government purposes notwithstanding any copyright notation herein.

REFERENCES

[1] U. Hoffmann, J.-M. Vesin, T. Ebrahimi, and K. Diserens, "An efficient P300-based brain–computer interface for disabled subjects," *Journal of Neuroscience methods,* vol. 167, no. 1, pp. 115-125, 2008.

[2] A. Jackson, C. T. Moritz, J. Mavoori, T. H. Lucas, and E. E. Fetz, "The Neurochip BCI: towards a neural prosthesis for upper limb function," *IEEE Transactions on Neural Systems and Rehabilitation Engineering,* vol. 14, no. 2, pp. 187-190, 2006.

[3] Y. Li, J. Pan, F. Wang, and Z. Yu, "A hybrid BCI system combining P300 and SSVEP and its application to wheelchair control," *IEEE Transactions on Biomedical Engineering,* vol. 60, no. 11, pp. 3156-3166, 2013.

[4] T. Nayak *et al.*, "Prediction of Human Performance Using Electroencephalography under Different Indoor Room Temperatures," *Brain sciences,* vol. 8, no. 4, p. 74, 2018.

[5] G. R. Muller-Putz and G. Pfurtscheller, "Control of an electrical prosthesis with an SSVEP-based BCI,"

IEEE Transactions on Biomedical Engineering, vol. 55, no. 1, pp. 361-364, 2007.

[6] L. A. Farwell and E. Donchin, "Talking off the top of your head: toward a mental prosthesis utilizing event-related brain potentials," *Electroencephalography and clinical Neurophysiology,* vol. 70, no. 6, pp. 510-523, 1988.

[7] J. Jin, E. W. Sellers, S. Zhou, Y. Zhang, X. Wang, and A. Cichocki, "A P300 brain–computer interface based on a modification of the mismatch negativity paradigm," *International journal of neural systems,* vol. 25, no. 03, p. 1550011, 2015.

[8] X. Chen, Y. Wang, M. Nakanishi, X. Gao, T.-P. Jung, and S. Gao, "High-speed spelling with a noninvasive brain–computer interface," *Proceedings of the national academy of sciences,* vol. 112, no. 44, pp. E6058-E6067, 2015.

[9] L. Meriño *et al.*, "Asynchronous control of unmanned aerial vehicles using a steady-state visual evoked potential-based brain computer interface," *Brain-Computer Interfaces,* vol. 4, no. 1-2, pp. 122-135, 2017.

[10] E. Yin, Z. Zhou, J. Jiang, Y. Yu, and D. Hu, "A dynamically optimized SSVEP brain–computer interface (BCI) speller," *IEEE Transactions on Biomedical Engineering,* vol. 62, no. 6, pp. 1447-1456, 2014.

[11] L. M. Alonso-Valerdi, R. A. Salido-Ruiz, and R. A. Ramirez-Mendoza, "Motor imagery based brain–computer interfaces: An emerging technology to rehabilitate motor deficits," *Neuropsychologia,* vol. 79, pp. 354-363, 2015.

[12] N. Birbaumer, "Breaking the silence: brain–computer interfaces (BCI) for communication and motor control," *Psychophysiology,* vol. 43, no. 6, pp. 517-532, 2006.

[13] Z. Wang, Y. Yu, M. Xu, Y. Liu, E. Yin, and Z. Zhou, "Towards a hybrid BCI gaming paradigm based on motor imagery and SSVEP," *International Journal of Human–Computer Interaction,* vol. 35, no. 3, pp. 197-205, 2019.

[14] A. Finke, A. Lenhardt, and H. Ritter, "The MindGame: a P300-based brain–computer interface game," *Neural Networks,* vol. 22, no. 9, pp. 1329-1333, 2009.

[15] R. Parafita, G. Pires, U. Nunes, and M. Castelo-Branco, "A spacecraft game controlled with a brain-computer interface using SSVEP with phase tagging," in *2013 IEEE 2nd international conference on serious games and applications for health (SeGAH)*, 2013, pp. 1-6: IEEE.

[16] N. Chumerin, N. V. Manyakov, M. van Vliet, A. Robben, A. Combaz, and M. M. Van Hulle, "Steady-state visual evoked potential-based computer gaming on a consumer-grade EEG device," *IEEE transactions on computational intelligence and ai in games,* vol. 5, no. 2, pp. 100-110, 2013.

[17] T. Minka, "Estimating a Dirichlet distribution," ed: Technical report, MIT, 2000.

[18] G. McArthur, T. Budd, and P. Michie, "The attentional blink and P300," *Neuroreport,* vol. 10, no. 17, pp. 3691-3695, 1999.

[19] M. Mostavi, S. Salekin, and Y. Huang, "Deep-2'-O-Me: Predicting 2'-O-methylation sites by Convolutional Neural Networks," in *2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2018, pp. 2394-2397: IEEE.

[20] M. Mostavi, Y.-C. Chiu, Y. Huang, and Y. Chen, "Convolutional neural network models for cancer type prediction based on gene expression," *arXiv preprint arXiv:1906.07794,* 2019.

[21] M. M. Hasib, T. Nayak, and Y. Huang, "A hierarchical LSTM model with attention for modeling EEG non-stationarity for human decision prediction," in *2018 IEEE EMBS international conference on biomedical & health informatics (BHI)*, 2018, pp. 104-107: IEEE.